

Data Model Summary Report

1. Project Overview

Project Name	DAX Depo
Student Name	Ravindra Kumar
GR ID	11638
Tool Used	Microsoft Power BI Desktop
File	DAX_Depo.pbix
Report Page	Matrix_Visuals (Page 1)
Total Tables	7 (2 Fact + 3 Dimension + 1 Measures Table)
Total Visuals	6 Matrix (Pivot) Tables
DAX Measures	11 Measures in 'Measures (2)' table
Schema Type	Star Schema

This project demonstrates advanced DAX (Data Analysis Expressions) in Power BI by building calculated measures grouped in a dedicated Measures table. The model uses a Star Schema with Fact and Dimension tables connected through active relationships. Matrix visuals are used to verify measure outputs across Region, Date, Sales Category, and Customer Segment dimensions.

2. Tables in the Data Model

Table Name	Type	Key Column	Notable Columns / Purpose
Sales_Fact	Fact Table	SalesID	SalesID, RegionID, ProductID, CustomerID, DateKey, Sales_Category, Revenue, Cost, Quantity
Returns_Fact	Fact Table	ReturnID	ReturnID, SalesID, ReturnDateKey, Reason
Customer_Dim	Dimension Table	CustomerID	CustomerID, FullName, Segment (Gold/Silver/Platinum)
Date_Dim	Dimension Table	DateKey	DateKey, Date, MonthName, Quarter, Year, FiscalYear
Region_Dim	Dimension Table	RegionID	RegionID, RegionName, Country, State, City
Product_Dim	Dimension Table	ProductID	ProductID, ProductName, Category, Subcategory, Brand
Measures (2)	Measures Table	—	Dedicated table housing all 11 DAX measures; no rows

3. Relationships & Cardinality

From Table (Dim)	Key Column	To Table (Fact)	Cardinality	Cross-Filter	Status
Customer_Dim	CustomerID	Sales_Fact	1 : *	Single	Active

Product_Dim	ProductID	Sales_Fact	1 : *	Single	Active
Region_Dim	RegionID	Sales_Fact	1 : *	Single	Active
Date_Dim	DateKey	Sales_Fact	1 : *	Single	Active
Date_Dim	DateKey	Returns_Fact	1 : *	Single	Inactive
Returns_Fact	SalesID	Sales_Fact	* : *	Single	Reference

Inactive Relationship Note: The Date_Dim → Returns_Fact relationship on DateKey is kept inactive to avoid ambiguity. It can be activated in DAX using **USERELATIONSHIP()** inside calculated measures (e.g., North_2023_Sales uses this pattern).

4. DAX Measures (Measures Table)

All measures are stored in the **Measures (2)** table — a best-practice approach for organizing calculated logic separately from raw data tables. The 11 measures span aggregation, filtering, time-intelligence, and conditional logic categories.

Measure Name	Category	DAX Function(s) Used	Description
Total_Sales	Aggregation	SUM()	Sum of all sales revenue
Total_Items	Aggregation	SUM() / COUNT()	Total quantity of items sold
Total_Cost	Aggregation	SUM()	Total cost across all sales
Total_Profit	Aggregation	SUM() / subtraction	Revenue minus cost (profit)
Total_Sales_ALL	Filter Removal	CALCULATE(), ALL()	Grand total ignoring all filters
Sales_% of Total	Ratio	DIVIDE(), ALL()	Each row's share of grand total
North_Sales	Filter Context	CALCULATE(), FILTER()	Sales filtered to North region only
Sales_Above_Avg	Conditional	CALCULATE(), AVERAGEX(), FILTER()	Sales only where revenue > average
High_Value_Sales	Conditional	CALCULATE(), FILTER()	Sales above a defined high-value threshold
North_2023_Sales	Multi-Filter	CALCULATE(), USERELATIONSHIP()	North region sales for FY2023 only
YTD_Sales	Time Intelligence	TOTALYTD() / DATESYTD()	Year-to-date cumulative sales
Last_6_Months_Sales	Time Intelligence	DATESINPERIOD(), LASTDATE()	Rolling 6-month sales window
YoY_Growth%	Time Intelligence	DIVIDE(), SAMEPERIODLASTYEAR()	Year-over-year growth percentage

5. Matrix Table Verification (Report: Matrix_Visuals)

Six Matrix (Pivot Table) visuals were created on the **Matrix_Visuals** page to verify the correctness and behavior of all DAX measures across different filter contexts.

Visual 1 — Sales by Region (ALL & Percentage)

Rows:	Region_Dim[RegionName]
Columns:	(None — single column layout)
Values:	Total_Sales Total_Sales_ALL Sales_% of Total
Tables:	Region_Dim, Sales_Fact, Measures (2)
Purpose:	Verifies CALCULATE + ALL() to show grand total alongside filtered totals, and DIVIDE() for share percentages.

Visual 2 — Region Comparison with Conditional Measures

Rows:	Region_Dim[RegionName]
Columns:	(None)
Values:	Total_Sales North_Sales Sales_Above_Avg High_Value_Sales
Tables:	Region_Dim, Sales_Fact, Measures (2)
Purpose:	Verifies FILTER() context transitions — North_Sales locks to North region; Sales_Above_Avg and High_Value_Sales apply conditional logic.

Visual 3 — Region with Time-Filtered & YTD Measures

Rows:	Region_Dim[RegionName]
Columns:	(None)
Values:	Total_Sales Total_Sales_ALL North_2023_Sales YTD_Sales
Tables:	Region_Dim, Date_Dim, Sales_Fact, Measures (2)
Purpose:	Tests USERELATIONSHIP() for the inactive Date→Returns path, and TOTALYTD() for cumulative year-to-date figures.

Visual 4 — Monthly Sales with Time Intelligence

Rows:	Date_Dim[MonthName]
Columns:	(None)
Values:	Total_Sales YTD_Sales Last_6_Months_Sales YoY_Growth%
Tables:	Date_Dim, Sales_Fact, Measures (2)
Purpose:	Full time-intelligence verification: YTD accumulation, rolling 6-month window with DATESINPERIOD, and year-over-year comparison using SAMEPERIODLASTYEAR.

Visual 5 — Sales by Category (Cost & Items)	
Rows:	Sales_Fact[Sales_Category]
Columns:	(None)
Values:	Total_Sales Total_Items Sales_% of Total Total_Cost
Tables:	Sales_Fact, Measures (2)
Purpose:	Validates aggregation measures on the fact table itself using Sales_Category as row context — tests SUM of items and cost.

Visual 6 — Customer Segment Profitability	
Rows:	Customer_Dim[Segment]
Columns:	(None)
Values:	Total_Sales Total_Items Total_Profit High_Value_Sales
Tables:	Customer_Dim, Sales_Fact, Measures (2)
Purpose:	Verifies profit calculation and high-value threshold across Gold, Silver, and Platinum segments via Customer_Dim relationship.

6. Challenges Faced & Solutions Applied

#	Challenge	Solution Applied
1	Inactive relationship between Date & Returns causing ambiguity	Kept OrderDate active; ReturnDate kept inactive (dotted). Used USERELATIONSHIP() in North_2023_Sales measure.
2	Bidirectional filters causing circular paths	All relationships set to Single cross-filter direction. Bidirectional used only for Many-to-Many cases.
3	Many-to-Many risk between Returns_Fact & Sales_Fact	Cardinality reviewed; adjusted join keys and used reference/bridge approach to avoid fan-out.
4	Organizing many DAX measures cleanly	Created a dedicated 'Measures (2)' table using Enter Data; all measures stored there for clean model view.
5	YoY measure requiring prior year date context	Used SAMEPERIODLASTYEAR() with DIVIDE() and BLANK() handling to avoid errors when prior year data is absent.
6	Rolling window measure across months	DATESINPERIOD() combined with LASTDATE() to create a dynamic 6-month trailing window for Last_6_Months_Sales.

7. Data Model Summary

Schema Type	Star Schema — 2 Fact tables, 4 Dimension tables, 1 Measures table
Total Tables	7 (Sales_Fact, Returns_Fact, Customer_Dim, Date_Dim, Region_Dim, Product_Dim, Measures (2))
Total Visuals	6 Matrix (Pivot Table) visuals on Matrix_Visuals page
DAX Measures	11 measures covering Aggregation, Filtering, Conditional, and Time Intelligence

Relationships	6 total — 4 Active (Dim→Fact), 1 Inactive (Date→Returns), 1 Reference (Returns→Sales)
Cross-Filter Default	Single (Dimension to Fact) to prevent circular dependencies
Time Intelligence	YTD, Last 6 Months, Year-over-Year Growth — all based on Date_Dim
Filter Measures	North_Sales, Sales_Above_Avg, High_Value_Sales — use CALCULATE() with FILTER()
Key Techniques	CALCULATE(), ALL(), FILTER(), USERELATIONSHIP(), TOTALYTD(), DATESINPERIOD(), SAMEPERIODLASTYEAR(), DIVIDE()
Report File	DAX_Depo.pbix Created from Cloud Power BI Release 2026.06